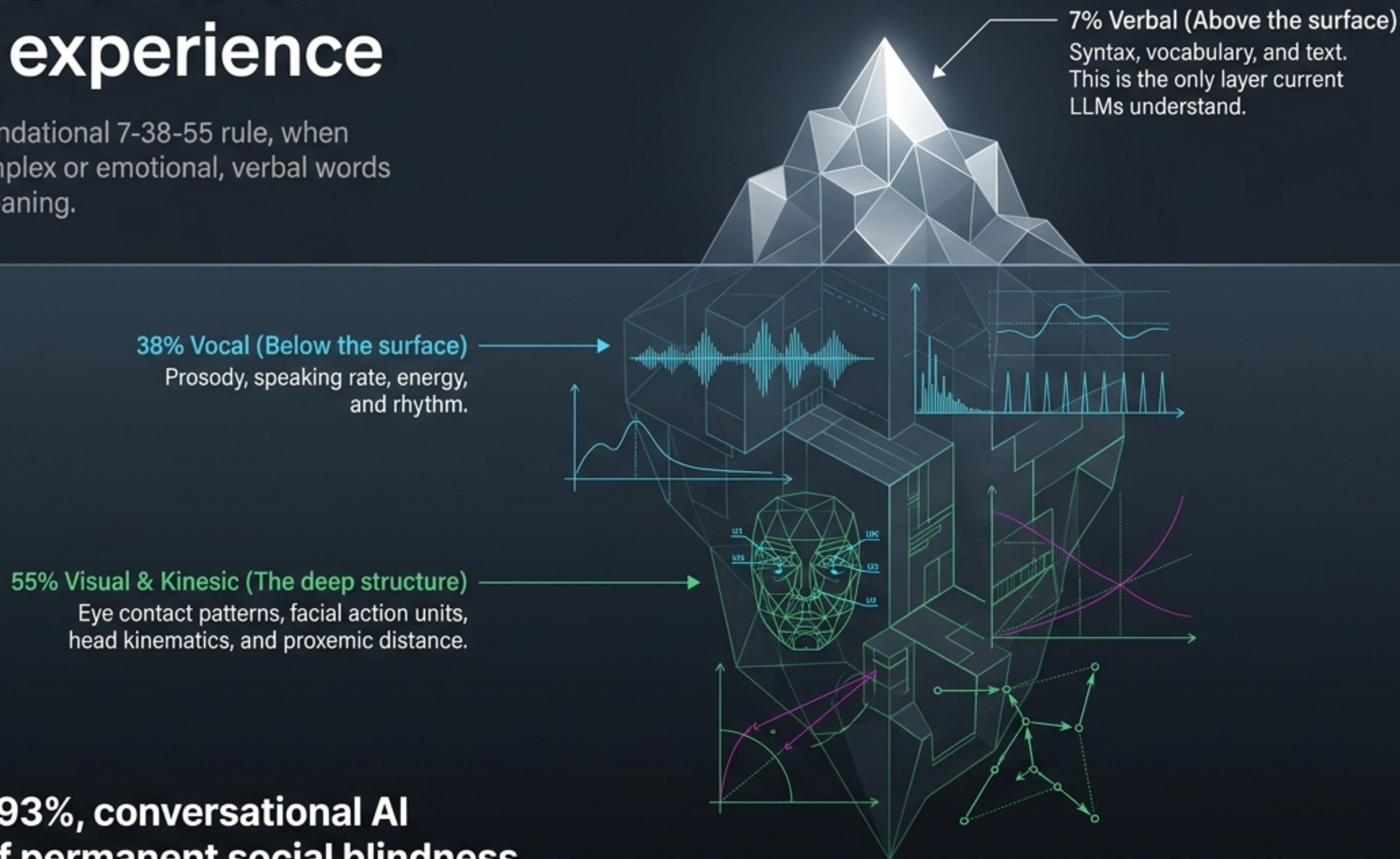


The Digital Social Nervus System

Engineering the missing 93% of human-computer interaction with HCEP v1.2.0.

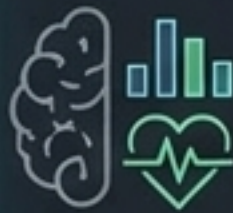
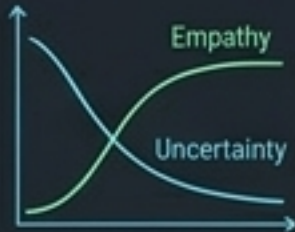
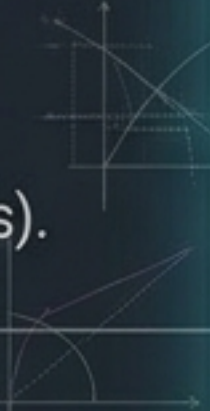
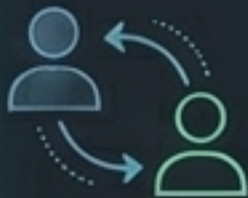
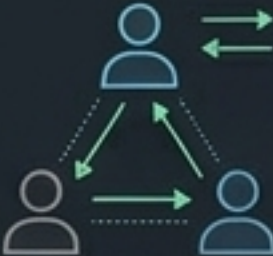
Modern AI is blind to the human experience

According to Mehrabian's foundational 7-38-55 rule, when human communication is complex or emotional, verbal words carry only a fraction of the meaning.



Without the bottom 93%, conversational AI operates in a state of permanent social blindness.

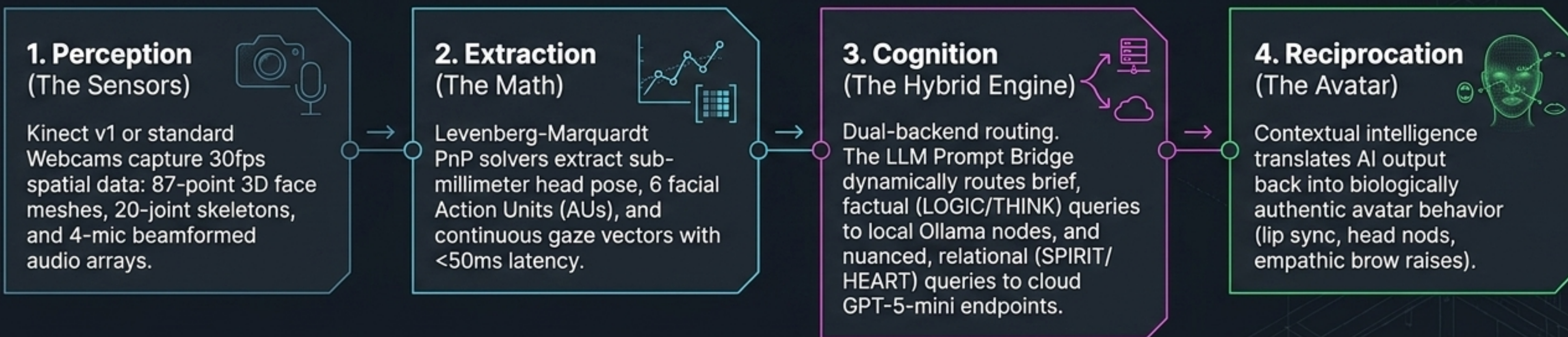
The Paradigm Shift in Human-Computer Interaction

Legacy AI Agents (Text & Voice)	Embodied HCEP Agents (Multimodal)
 Perception: Blind and asynchronous. Processes discrete text/audio inputs after the user finishes speaking. 	 Perception: Synchronous world-space awareness. Processes 30fps multi-modal streams (gaze, AUs, head kinematics). 
 Context: Semantic only. Understands the definitions of words but not the epistemic state of the speaker. 	 Context: Cognitive and Emotional. Evaluates uncertainty, processing load, and empathic states in real-time. 
 Expression: Scripted and linear. Avatars use randomized or audio-amplitude-driven animations (dead eyes). 	 Expression: Biologically authentic. Reciprocates with phoneme-accurate visemes, micro-saccades, and empathic mirror responses. 
 Interaction: Transactional. Waits for a prompt, generates a response. 	 Interaction: Relational. Engages in backchanneling, social triangle scanning, and knows when to remain silent. 

Human Communication Eye Protocol (HCEP)

A production-ready (v1.2.0) Social Signal Processing architecture that fuses sensor data with hybrid LLM reasoning to understand and reciprocate human communication.

Nervous System Flowchart

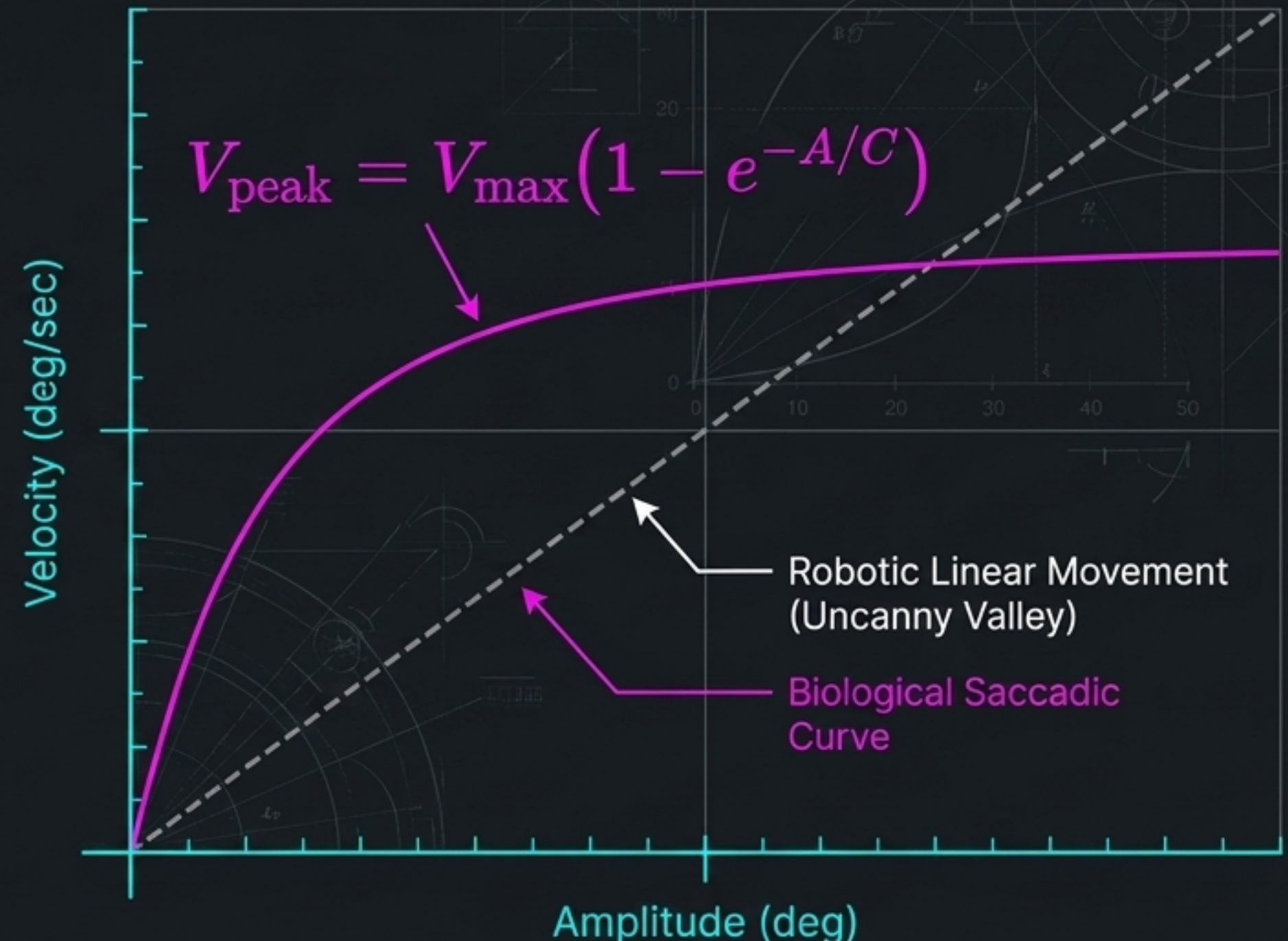


Engineering Biological Authenticity: The Saccadic Main Sequence

Current 3D avatars trigger the Uncanny Valley because their eyes move with linear, constant velocities. HCEP calculates human eye movement using the biological Saccadic Main Sequence.

HCEP replaces robotic panning with authentic saccadic kinematics:

- **Saccades:** Rapid jumps between fixation points, governed by amplitude-dependent velocity curves.
- **Micro-saccades:** 0.1-0.2° involuntary fixational movements that prevent receptor adaptation.
- **Blink Rhythm:** Biologically plausible intervals (3-8 per minute), modulated dynamically by the user's cognitive load and emotional state.



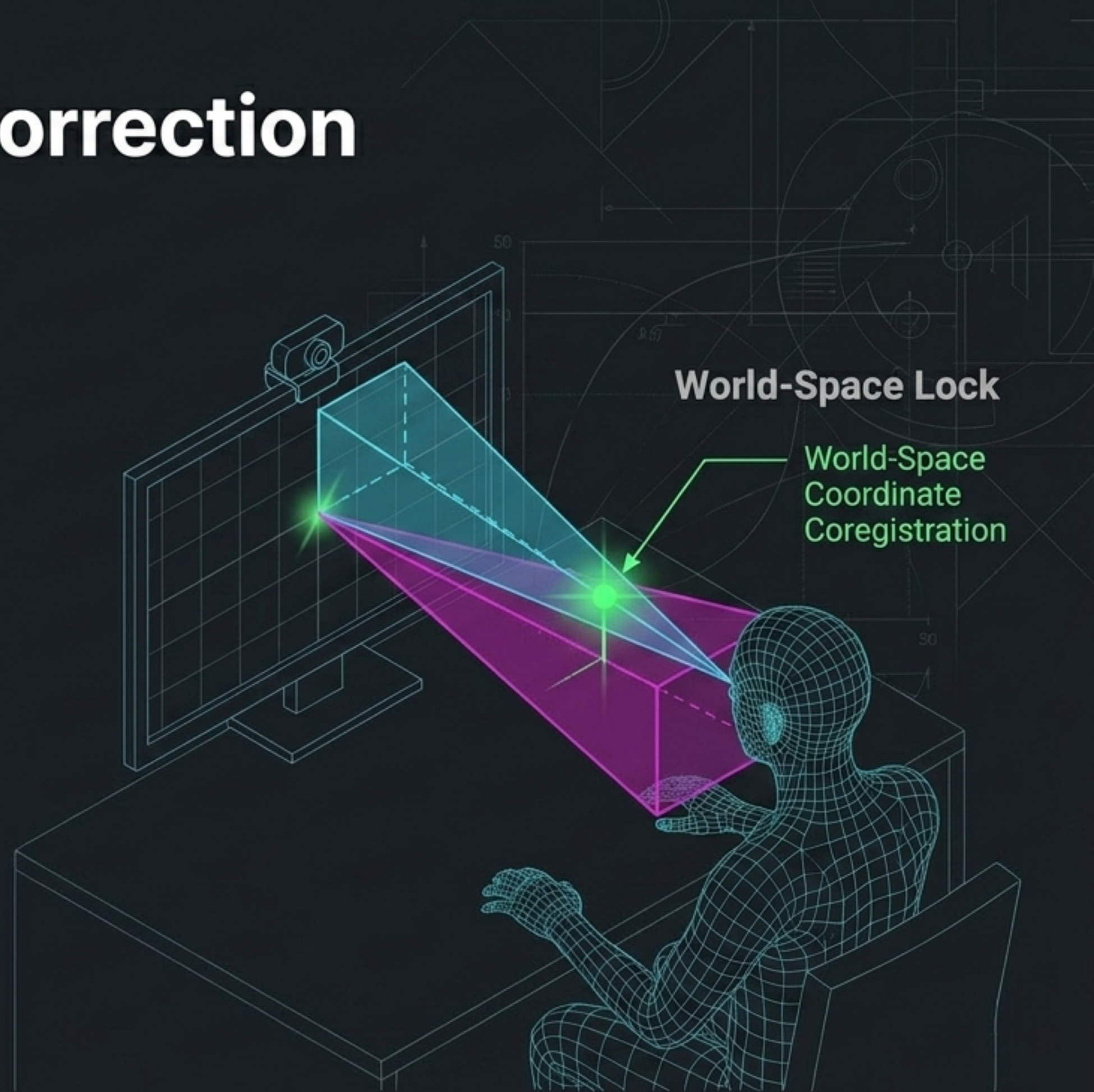
True Gaze™ Parallax Correction

Raw sensor gaze tracking fails in real-world deployments due to off-axis camera skew (e.g., a webcam mounted on top of a monitor bezel).

HCEP implements dynamic 3D coordinate co-registration to establish a World-Space Lock:

- Extracts 3D head translation ($X/Y/Z$) and Euler rotation angles ($Pitch/Yaw/Roll$).
- Applies a pinhole camera projection ($fx=fy=525$, $cx=320$, $cy=240$).
- Shifts the tracking perspective from the sensor's physical lens center directly to the active user-avatar focal line of sight.

Result: The AI avatar makes eye contact exactly where the user is looking on the physical screen.



The 5-Mode HCEP Classification

HCEP maps specific nonverbal behaviors to a continuous cognitive-emotional state machine, enabling the AI to adjust its tone and depth in real time.

Mode	Gaze Pattern	Key Action Units (AUs)	Cognitive State	AI Response Strategy
LOGIC	Structured, on-face	AU4 (frown) mild	Analytical	Precise, factual, numbered lists
AFFECT	Social Triangle (eyes + mouth)	AU12 > 0.2, AU6 > 0.1	Emotional engagement	Warm, empathetic, feeling-first
SPIRIT	Sustained mutual gaze (>3s)	Low activation, relaxed	Deep authentic rapport	Personal, genuine, unstructured
HEART	Lower-face emphasis	AU1+AU4, AU15	Empathic resonance	Supportive, validating, caring
THINK	Aversion (>15°), defocused	AU4, low AU12	Internal processing	Brief, non-intrusive, space-giving

Visualizing AFFECT: The Social Triangle

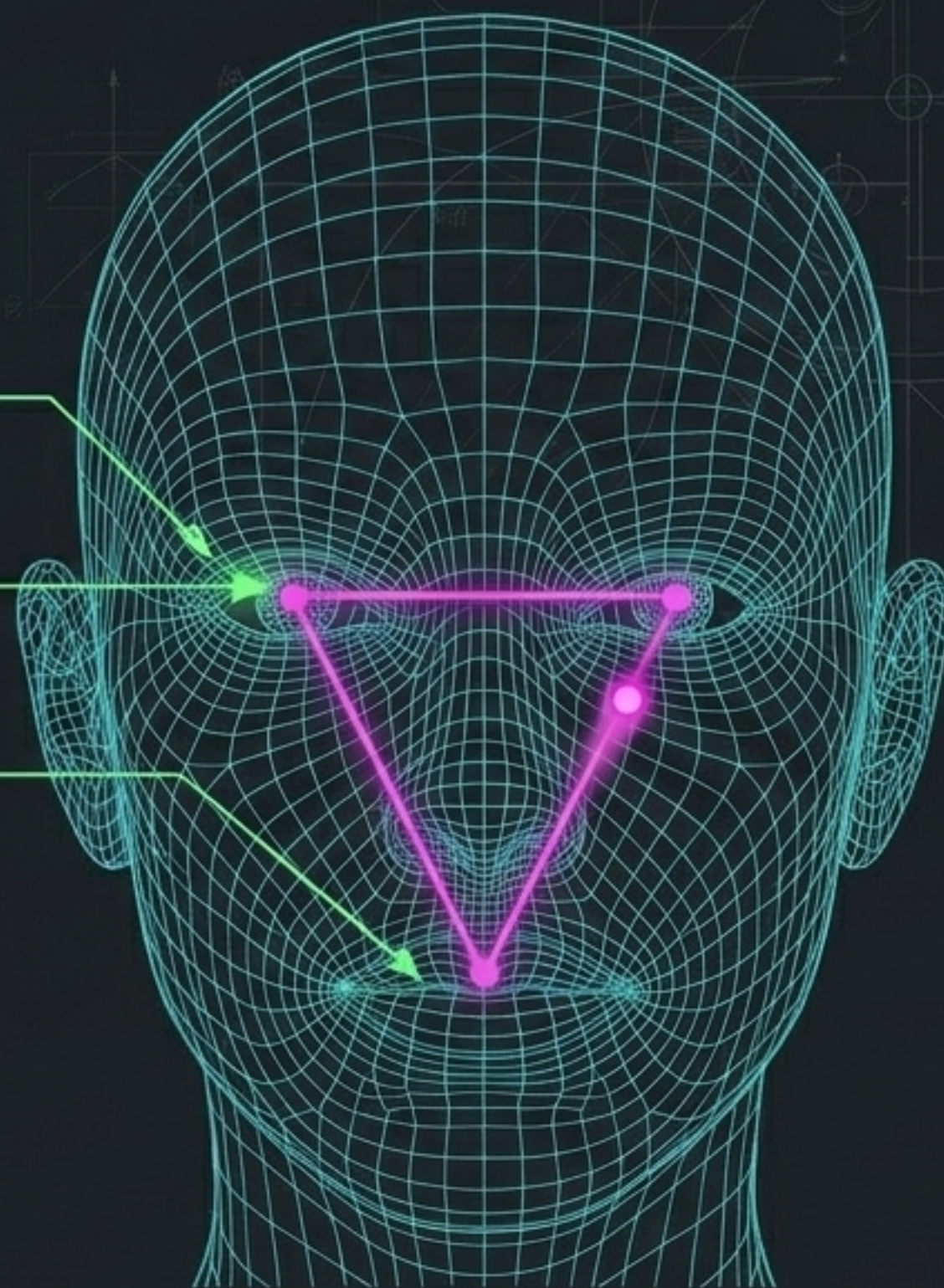
During positive social engagement, humans do not stare fixedly. They scan the Social Triangle in a systematic, rhythmic pattern (Argyle et al., 1973).

Left Eye: ~35-40% of fixation time.

Right Eye: ~35-40% of fixation time.

Mouth: ~15-20% of fixation time.

The HCEP Translation: When the system detects this specific geometric scanning cycle paired with the Duchenne marker (AU12 lip pull + AU6 cheek raise), HCEP locks into AFFECT mode. The LLM Prompt Bridge instantly routes the context to a cloud LLM instructed to respond with high-valence, high-arousal social warmth.

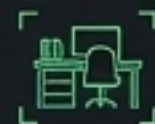
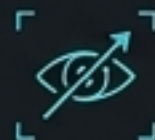


Contextual Intelligence & The Silence Protocol

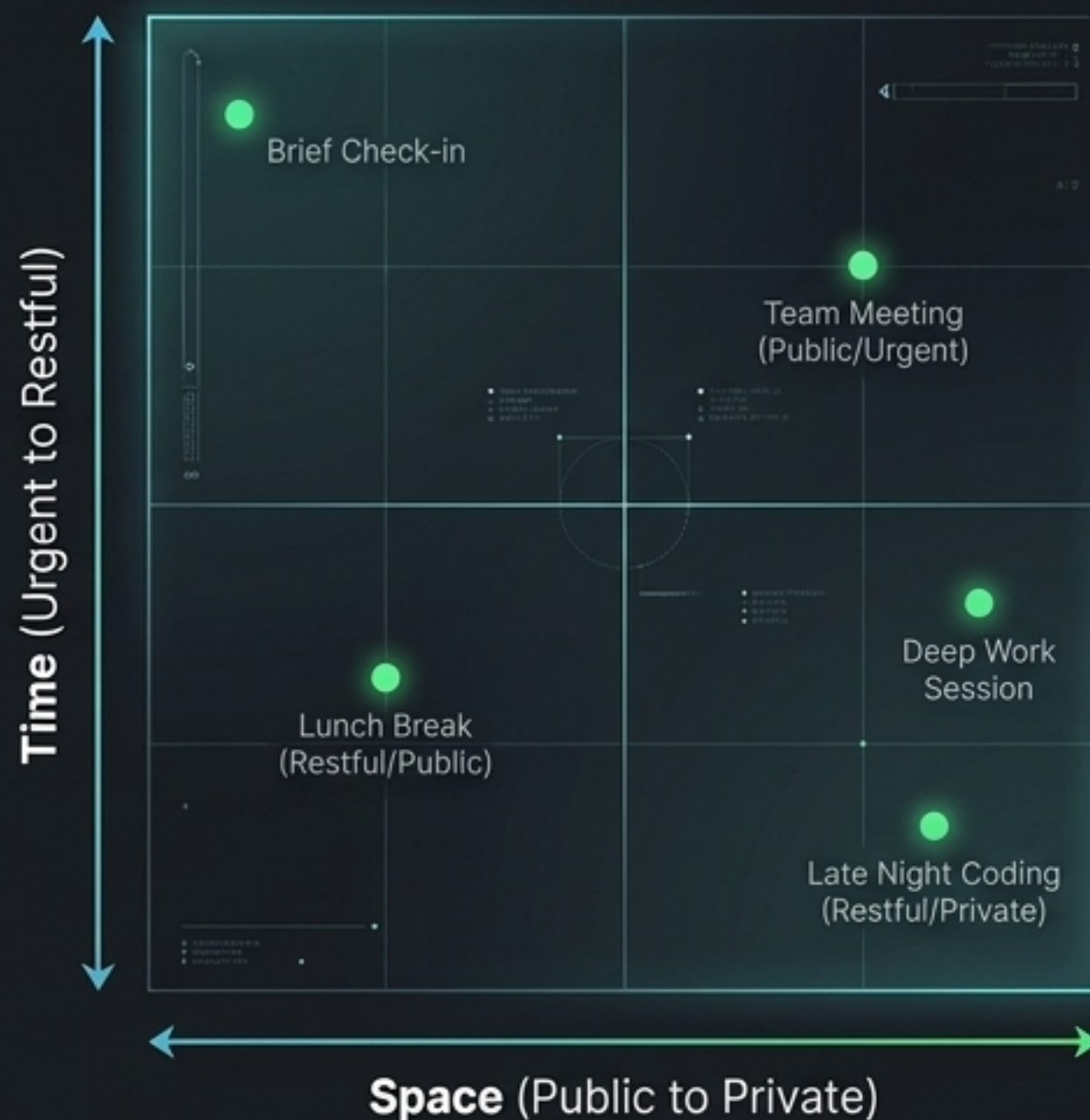
The most sophisticated communicative act for an AI is not speech—it is knowing when to be silent. HCEP's **ContextSnapshot** captures **Time × Space × Situation** to modulate behavior.

Speak-Suppression Signals (AI Remains Silent):

- **Processing Silence:** THINK mode + gaze aversion. The user is internally processing; any AI speech interrupts this.
- **Deep Work:** Sustained brow furrow (AU4 > 3s) + Laboratory/Office environment.
- **Empathic Presence:** HEART mode + late night (22:00+). Maximum attunement requires presence, not words.



22:00+



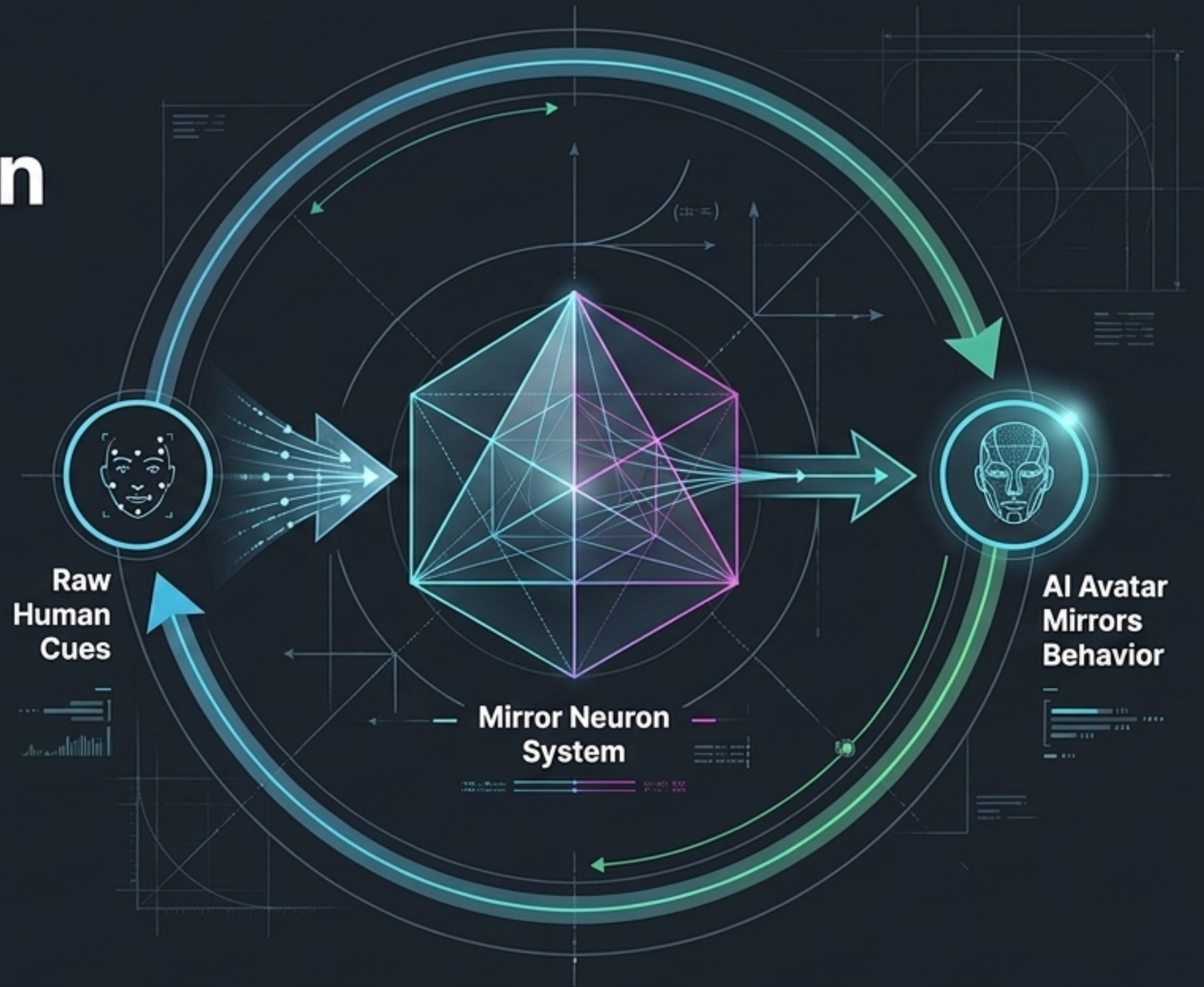
Turn-Yield Signals (AI Responds):

- **Floor Yield:** Direct, sustained gaze toward the avatar.
- **Question Signal:** Raised brows (AU1 + AU2) combined with direct gaze.



The Synthesis: Reciprocal Expression

An AI that only perceives is a surveillance tool. An AI that perceives and reciprocates is a social partner. HCEP creates a closed-loop digital nervous system based on human mirror neuron pathways.



- **Phoneme-Accurate Lip Sync (Phase 13):** Utilizes the 21-viseme SAPI taxonomy mapped to the Preston Blair animation canon. 60ms EMA co-articulation blending ensures mouth movements perfectly match auditory phonemes, curing the McGurk Effect intelligibility drop.

- **Empathic Mirroring:** Detects a human smile (AU12) and triggers a 200-400ms biological delay before generating an avatar micro-smile.

- **Backchannel Nodding:** Phase-locked head nods generated during human speech at prosodic boundaries to signal active listening.

Empirically Validated. Production Ready.

HCEP v1.2.0 is a production-hardened architecture built on .NET 9.0, utilizing bounded channels for backpressure-aware async data flow at a locked 30fps.

Validation Study Results

Based on 6,000 frames of conversational data evaluated by three independent expert raters.

0.8084

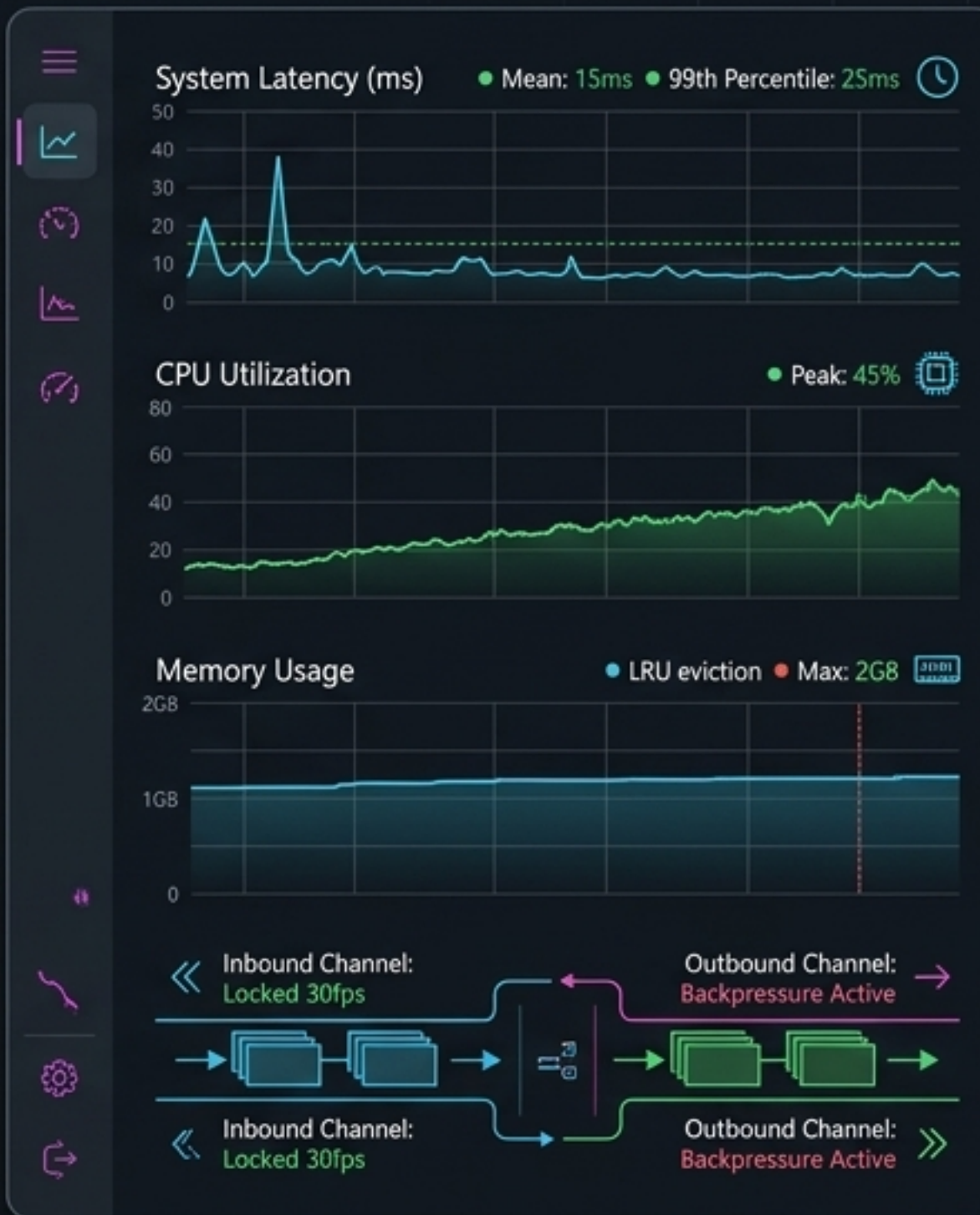
Inter-Rater Reliability (Cohen's Kappa):
Exceeds the ≥ 0.70 target, indicating
Almost Perfect Agreement.

84.55%

Classification Accuracy: The HCEP
pipeline accurately matches human
consensus on cognitive-emotional states.

Enterprise Hardening

- Cloud LLM circuit breakers with 30s failover cooling.
- DPAPI-encrypted Windows Credential Manager integration for API key security.
- In-memory knowledge store capacity limits with LRU eviction to prevent unbounded memory growth.

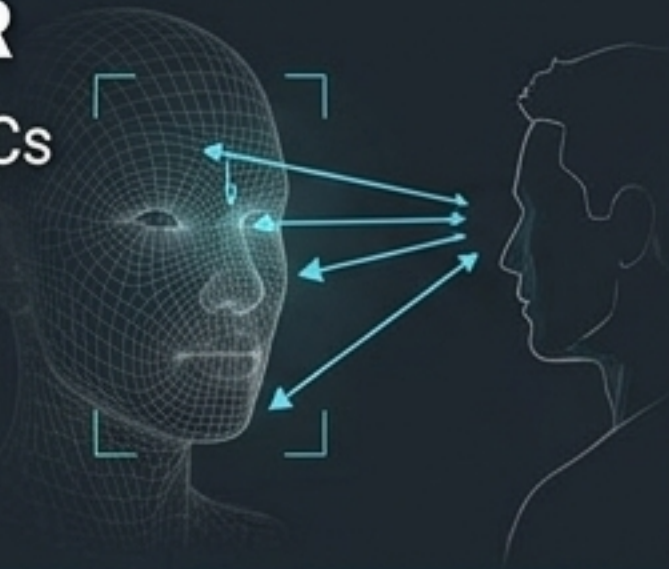


Transforming Human-Machine Interfaces

HCEP's multi-platform SDKs (Python, C#, Unity, Unreal Engine) enable immediate integration across critical sectors:

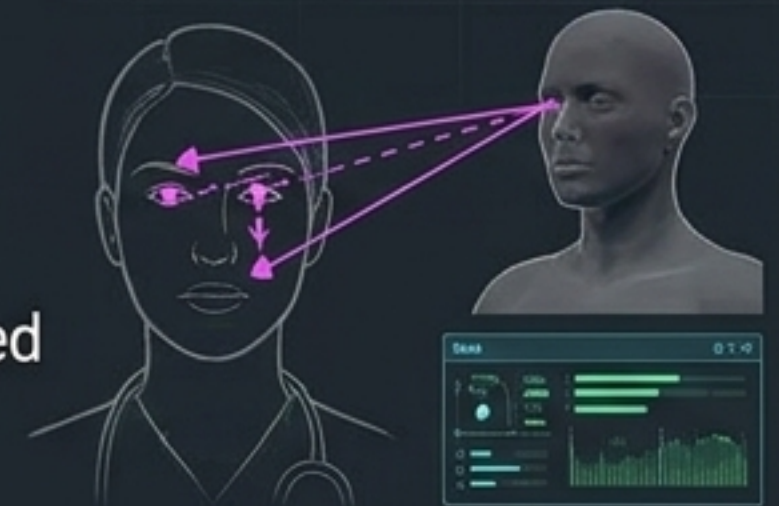
Game Engine NPCs & VR

Break the dead eyes barrier. NPCs that execute computational attention, scanning the player's face and responding to conversational dominance dynamics.



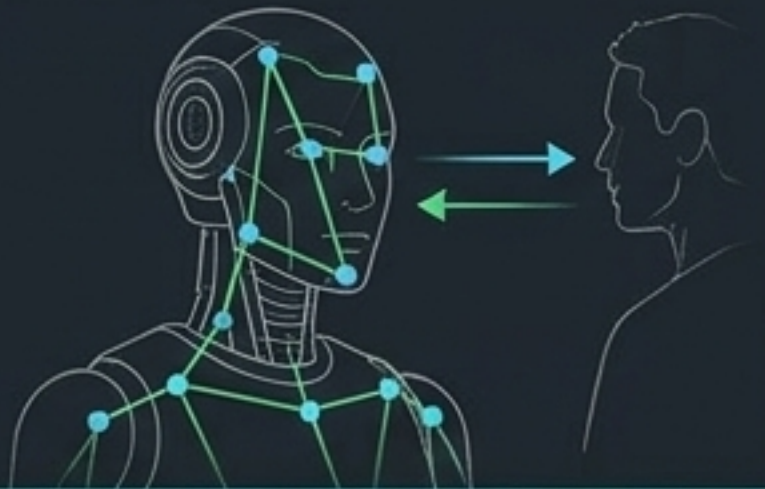
Medical Education

Provide objective, real-time gaze behavior analysis and feedback for medical students during standardized patient simulations.



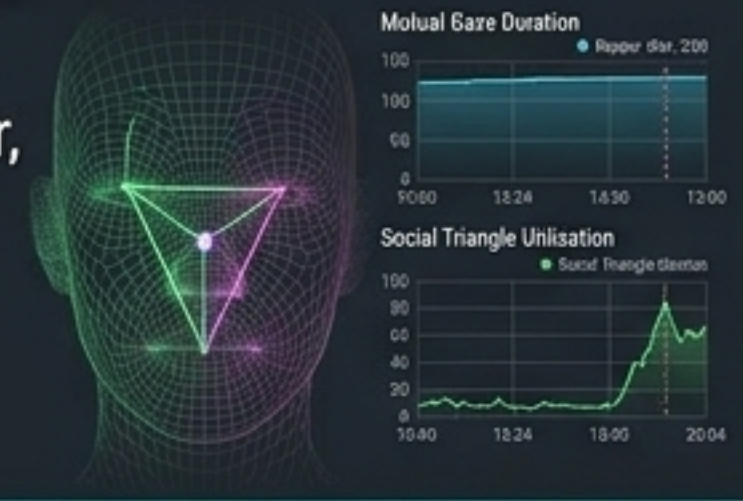
Companion Robotics (ROS2)

Equip physical humanoid robots with genuine social responsiveness, enabling therapeutic presence and joint attention capabilities.



Autism Therapy & Diagnostics

Serve as an objective behavioral biomarker monitor, tracking social triangle utilization and mutual gaze duration over longitudinal studies.



Mirror Neuron System

Mirroring & mirroring actions

Mirroring actions & intentions

The Social Turing Test

The original Turing Test proposed text-based conversation as the benchmark for **intelligence**. But human connection is 93% non-verbal.

The true benchmark for AI is the **Social Turing Test**: sustaining face-to-face interaction using gaze, expression, and gesture with such biological authenticity that a human feels truly seen, truly heard, and truly met.

HCEP is the foundational protocol to pass it.

Mirroring actions & intentions

Data flow system actions